Appendix 8: How to Parse and Crack the In-Vehicle CAN Protocol - DBC Protocol

Instruction Manual

Instruction Manual Version: V2.04

Update Date: 2023.07.01

DBC Protocol Parsing

Each CAN channel can independently support the parsing of vehicle CAN-bus application protocols, as long as the user imports the relevant DBC file in CANtest or CANpro1.50 software, the parsing of application layer data can be realized. It can be used for vehicle CAN protocol analysis, vehicle testing and maintenance, and cracking vehicle CAN protocol.

When users use CANTest or CANPro1.50, they only need to replace ControlCAN.dll and other library files (refer to: How to Compatible with the use of Zhou Ligong CANTest/CANpro1.50 software.pdf), and select the model number: USBCAN-2E-U can be.

CANpro1.50 is more feature-rich, here take CANpro1.50 as an example.

1、Operation Steps

Open the CANpro1.50 software, select the USBCAN-2E-U interface card, and select the baud rate of the bus (subject to the actual baud rate, the automotive CAN bus is generally 500K), click OK and start to start the CAN interface card. As shown in Figure 2;

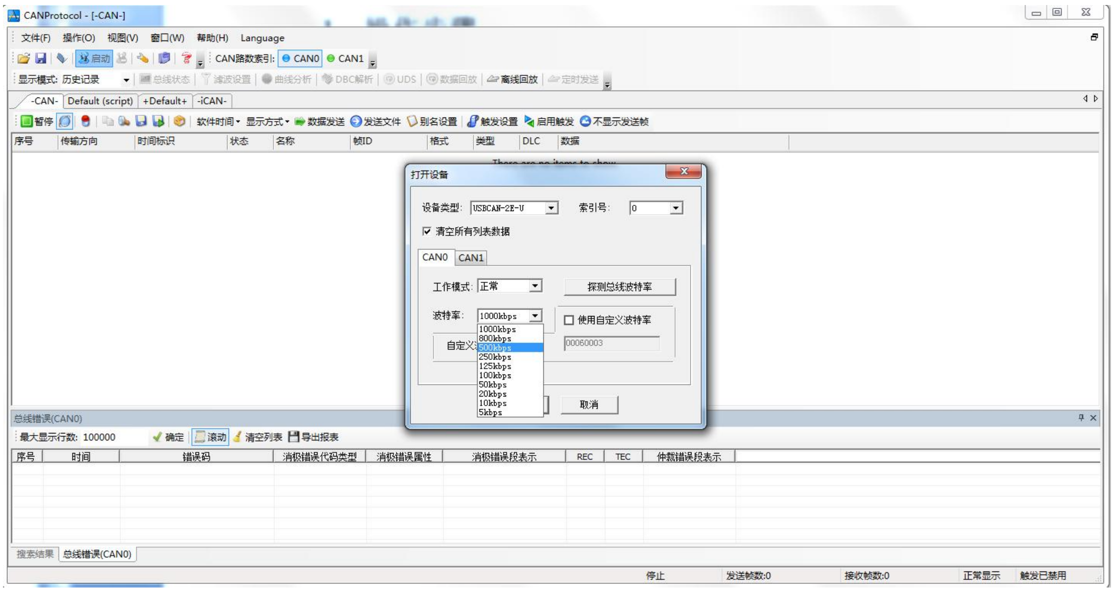


Figure 2 Open CANpro Software Initialization

Click the DBC Parsing button in the menu shortcut operation to enter the DBC Parsing interface, as shown in Figure 3;

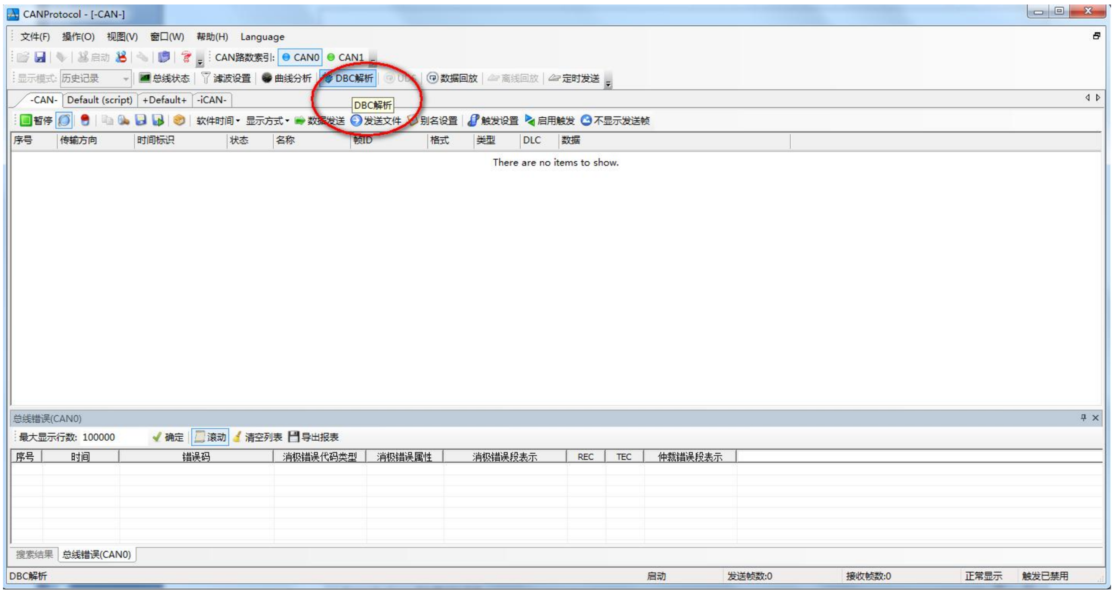


Figure 3 Open DBC Parsing

In the DBC parsing interface, click Load DBC and select the corresponding DBC file to open, CANpro comes with three DBC files. In this paper, take J1939 protocol as an example, select j1939.dbc to open it and parse the protocols of diesel engine, truck or bus, etc., as shown in Figure 4.

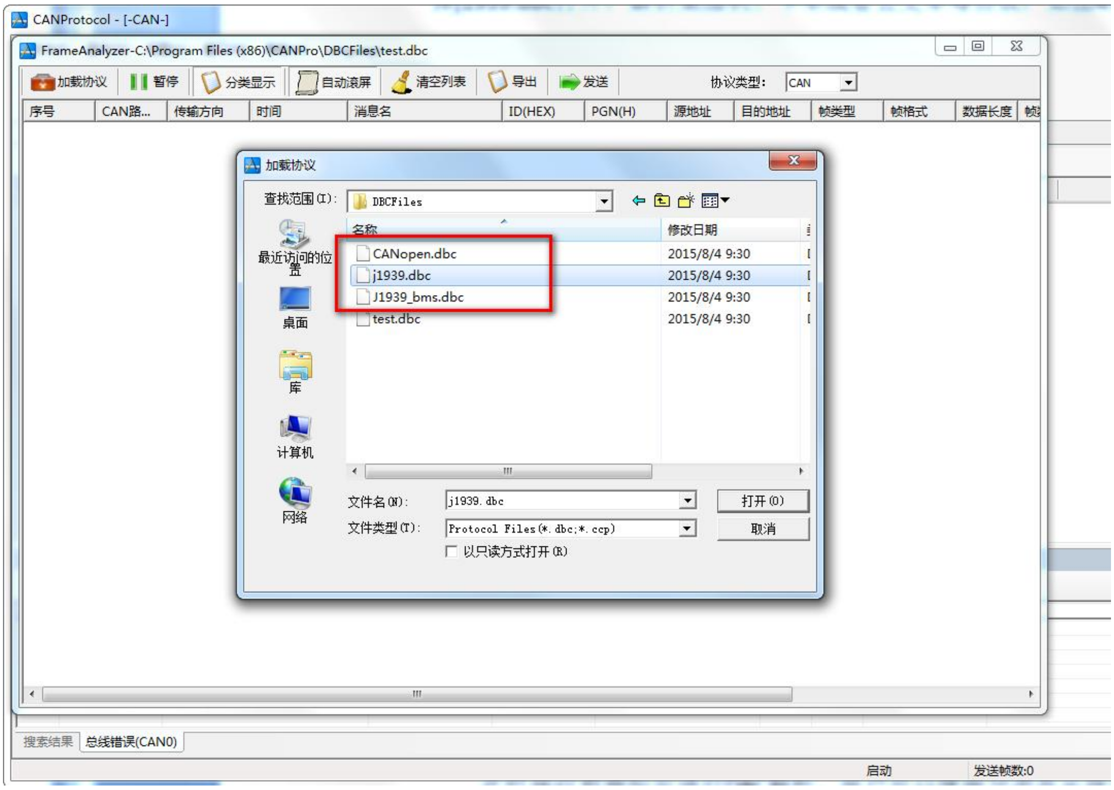


Figure 4 Load DBC File

At this point, the received data can be parsed by DBC, and the user can view it by sorting or refreshing the display. Click on a message, the application data contained in this frame will be displayed in the parsing box below. As shown in Figure 5, ID 0x0CF0041A in the fourth byte is 0x6C, the fifth byte is 0xD6. check, against the SAE_J1939-71 protocol: Electronic Engine Controller #1: EEC1 (message name) in the fourth and fifth bytes represent EngSpeed (engine speed).

Data length: 2 bytes

Resolution: 0.125 rpm/bit increments from 0 rpm (high byte resolution = 32 rpm/bit)

Data range: 0 to 8031.875 rpm

RPM can be calculated: 0xD66C*0.125 is 6861.50 rpm (revolutions per minute).

For the definition and resolution of other parameters, please refer to SAE_J1939-71 Protocol: CD-ROM\Instructions Documentation Table of Contents\16.Annex 9: SAE_J1939-71 Protocol.pdf

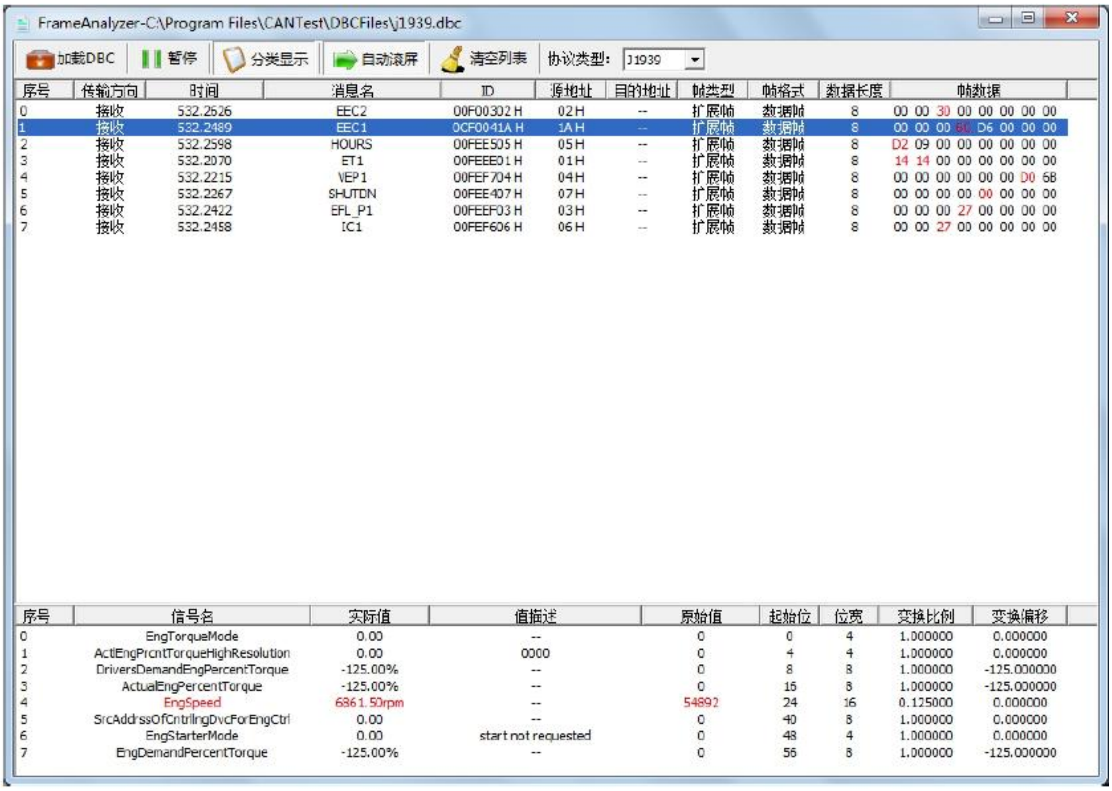
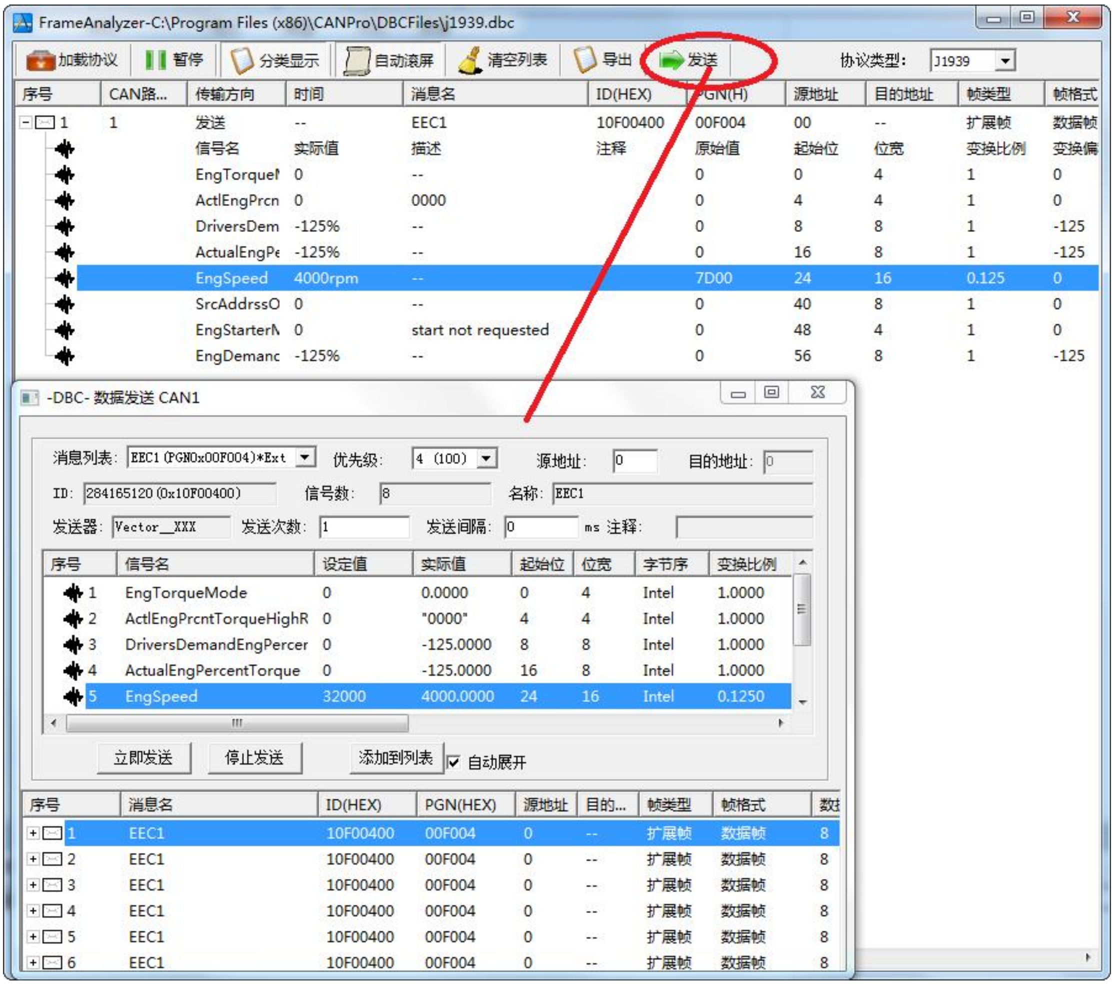


Figure 5 DBC Protocol Analysis Results

Tip: When using the classification display function, the software will mark the data with changes in red, which can help users to quickly complete the identification of variables when deciphering unknown protocols. For example, if you want to know the CANID and data segment corresponding to the steering wheel, you can use this method to run, turn the steering wheel, and observe the variables that turn red, that is, the corresponding.

The latest version of CANPro 1.50 software supports the DBC protocol sending function.

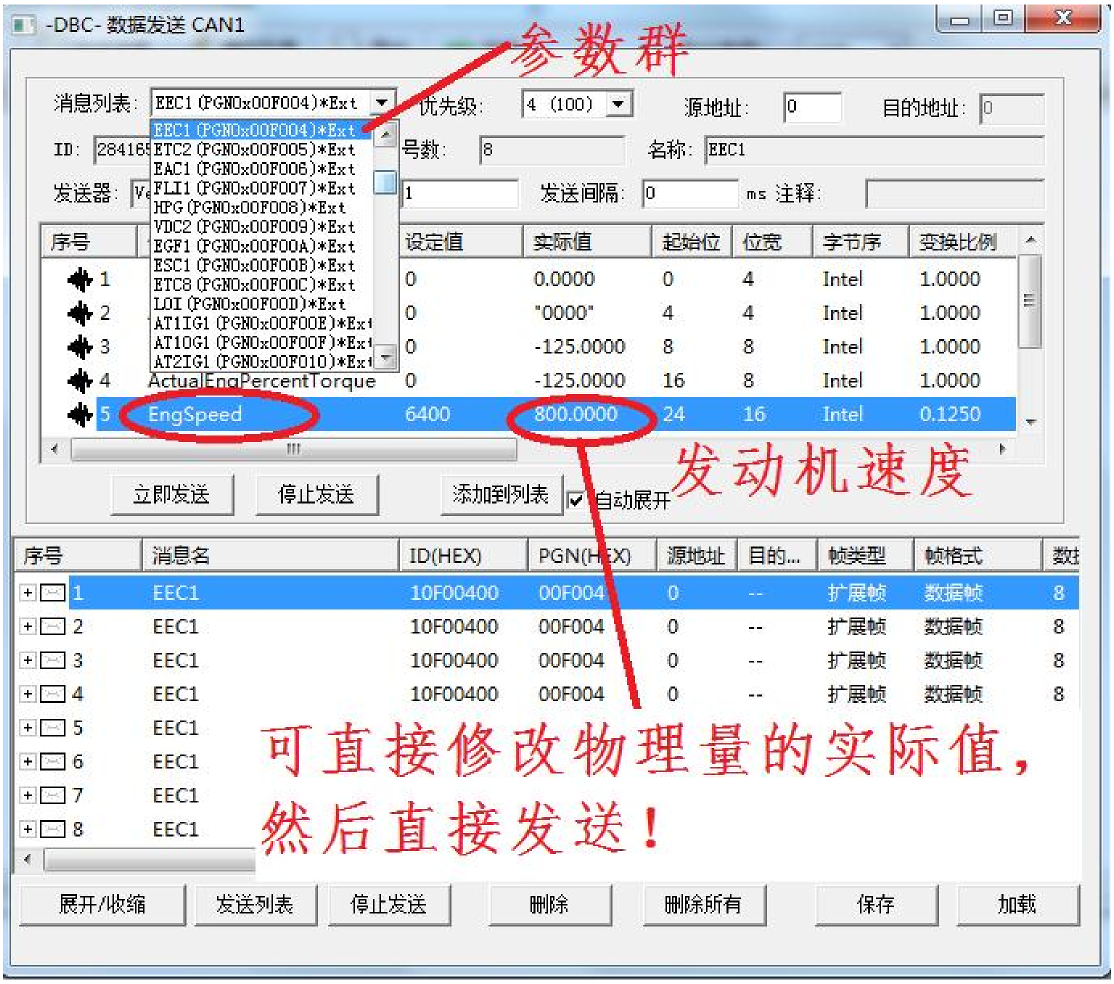


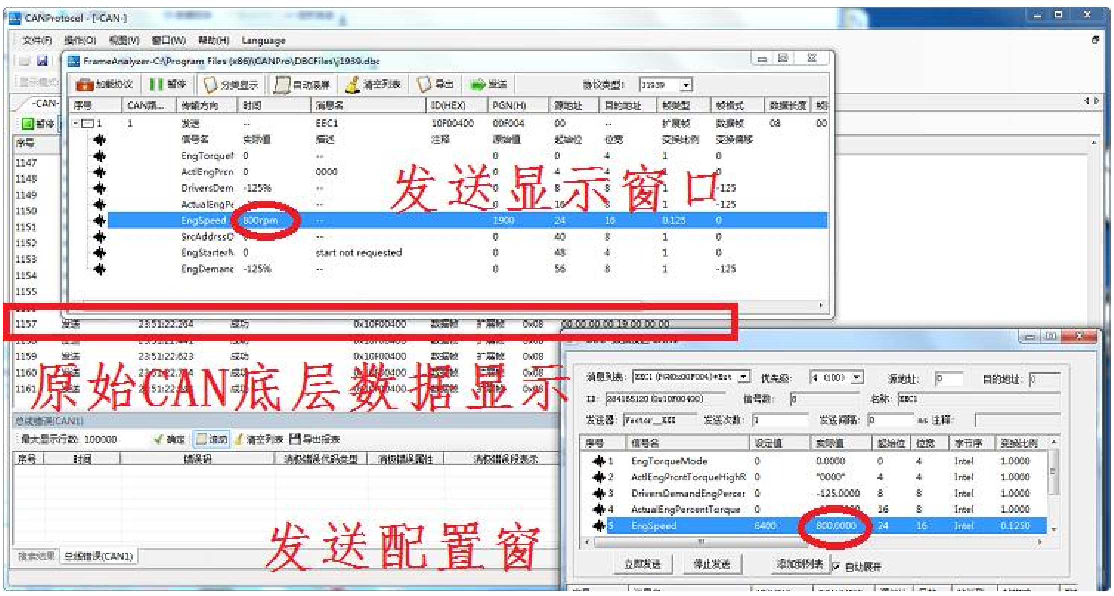
Supports visual configuration of sending data according to J1939 protocol.

J1939 protocol sending interface:

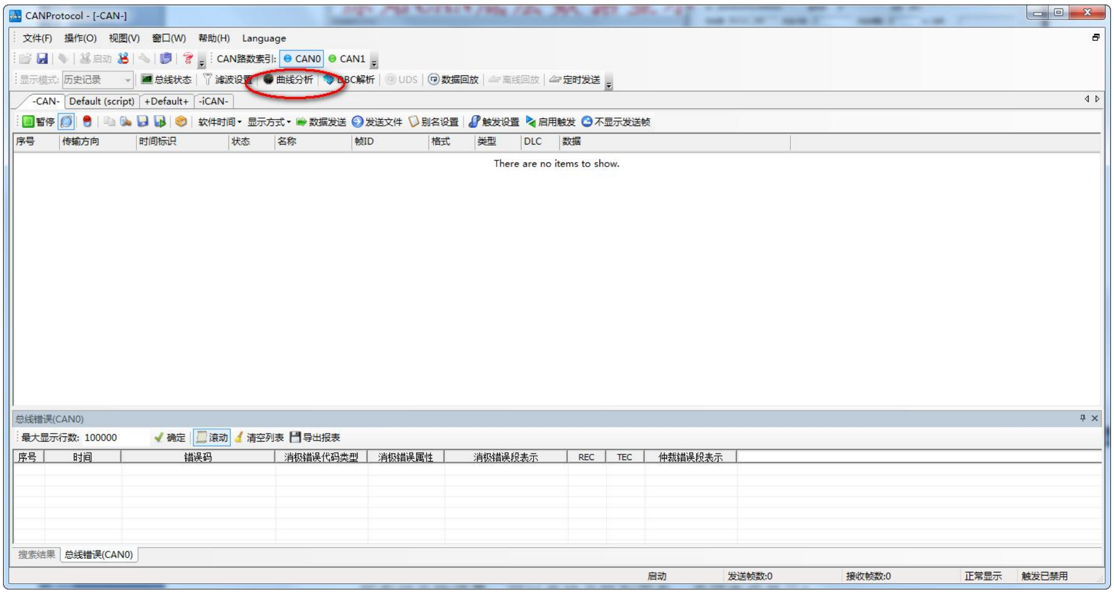
Any physical parameter of the parameter group is defined with actual value and sent. Ignore the principle of J1939 protocol. You can directly simulate the J1939 device for testing.

In the following figure, configure the simulated engine speed of 800 rpm and send.

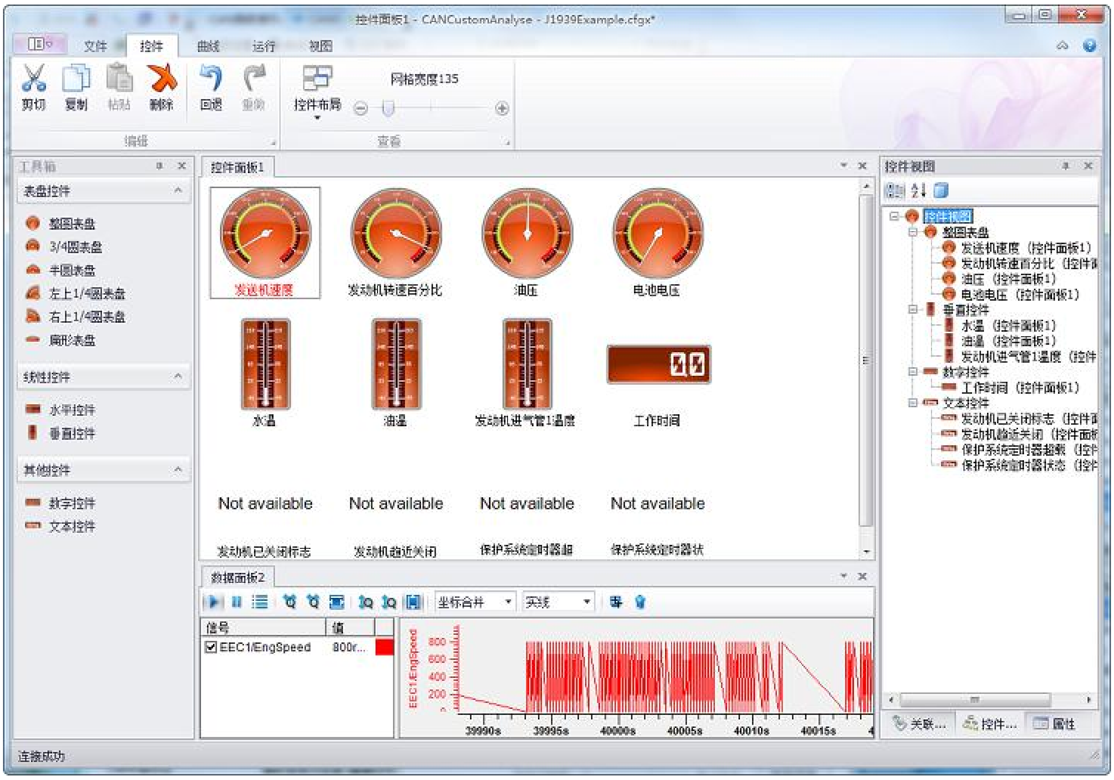




The latest version of CANPro 1.50 software supports DBC protocol curve and graph display.



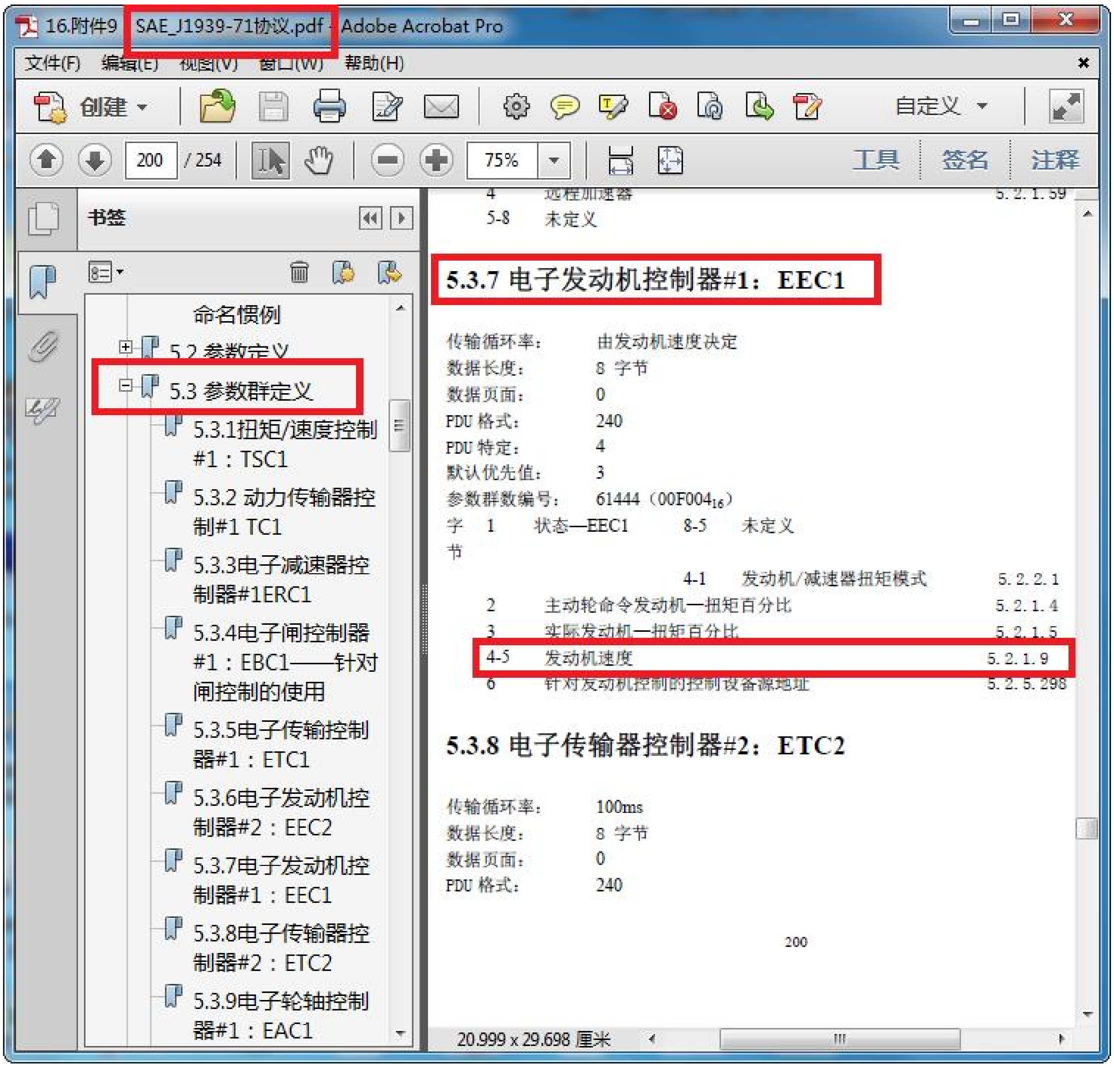
All relevant physical quantities can be directly associated to graphs and curves for visualization!

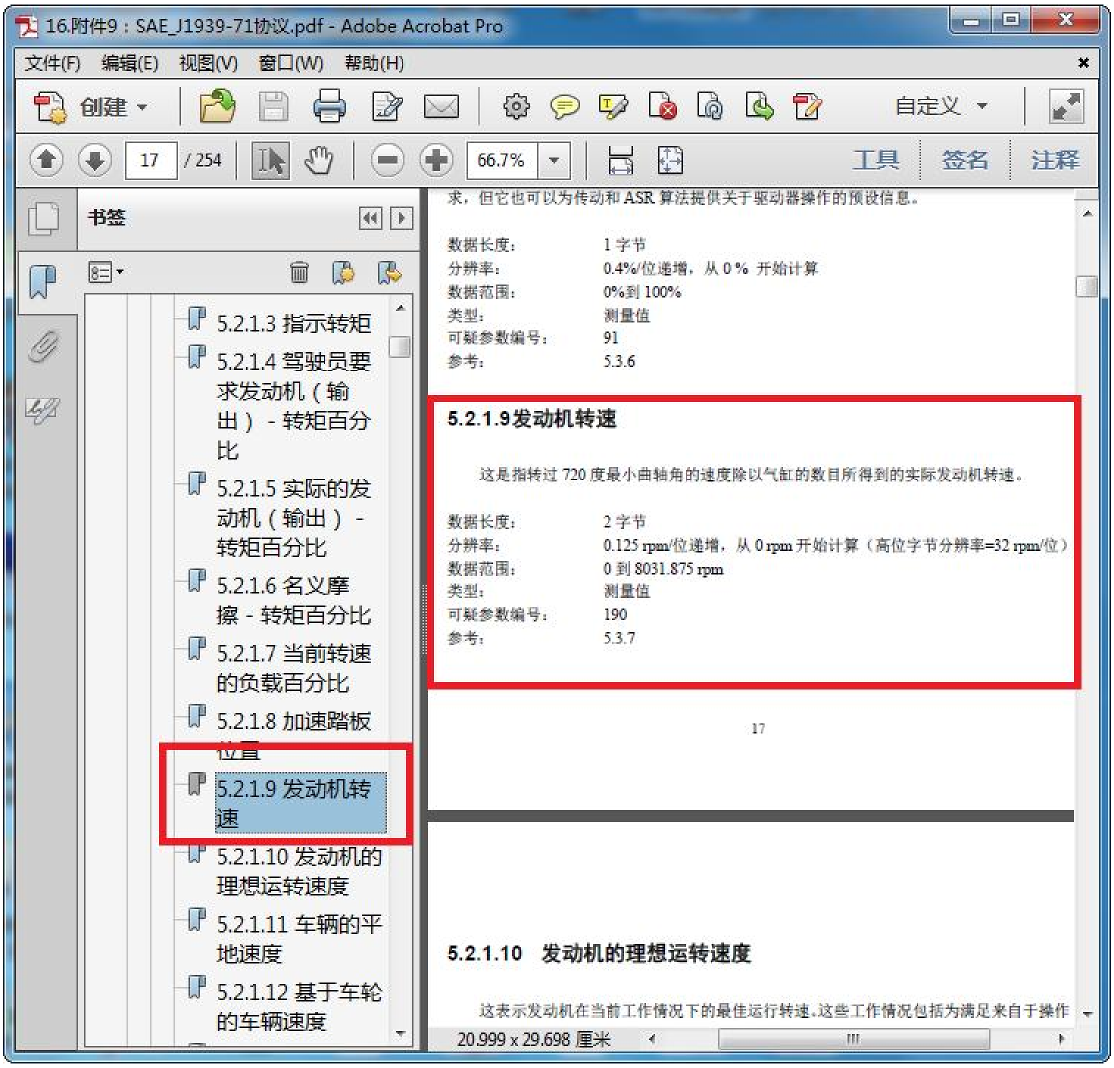


If you don't have a real car, you can also debug and learn the J1939 protocol, buy a dual-channel product, and you can send and receive data between the two channels.


CANPro - [-CAN-]
底层CAN数据列表
CAN0通道接收，
并解析J1939协议
发动机转速
EEC1/EngSpeed
CAN1通道按J1939协议
发送数据！
EngTorqueMode
ActlEngPrcntTorqueHighR
DriversDemandEngPercer
ActualEngPercentTorque
EngSpeed

The parameter groups in the software are displayed in English, and the names of the controls in the control panel can be modified to Chinese, and the Chinese version of the protocol is provided here for comparison.





2、Application Scope

- Industrial Control Testing

- Automotive Electronics Maintenance and Repair

- Protocol Cracking